

Lecture#1: Introduction to Knowledge Graphs and their Real-World Applications

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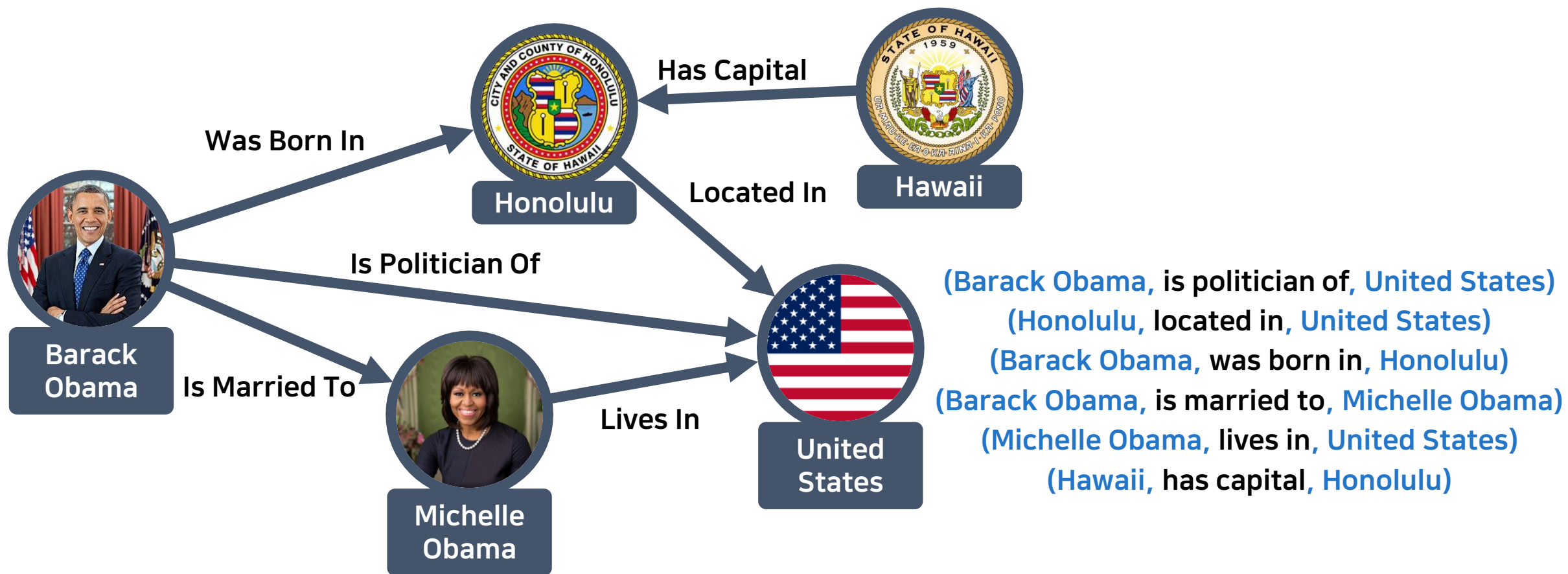
Knowledge Discovery with Knowledge Graphs: From Structural Embeddings to Generative Reasoning (KnowKG), KDD 2026 Tutorial

<https://bdi-lab.kaist.ac.kr>



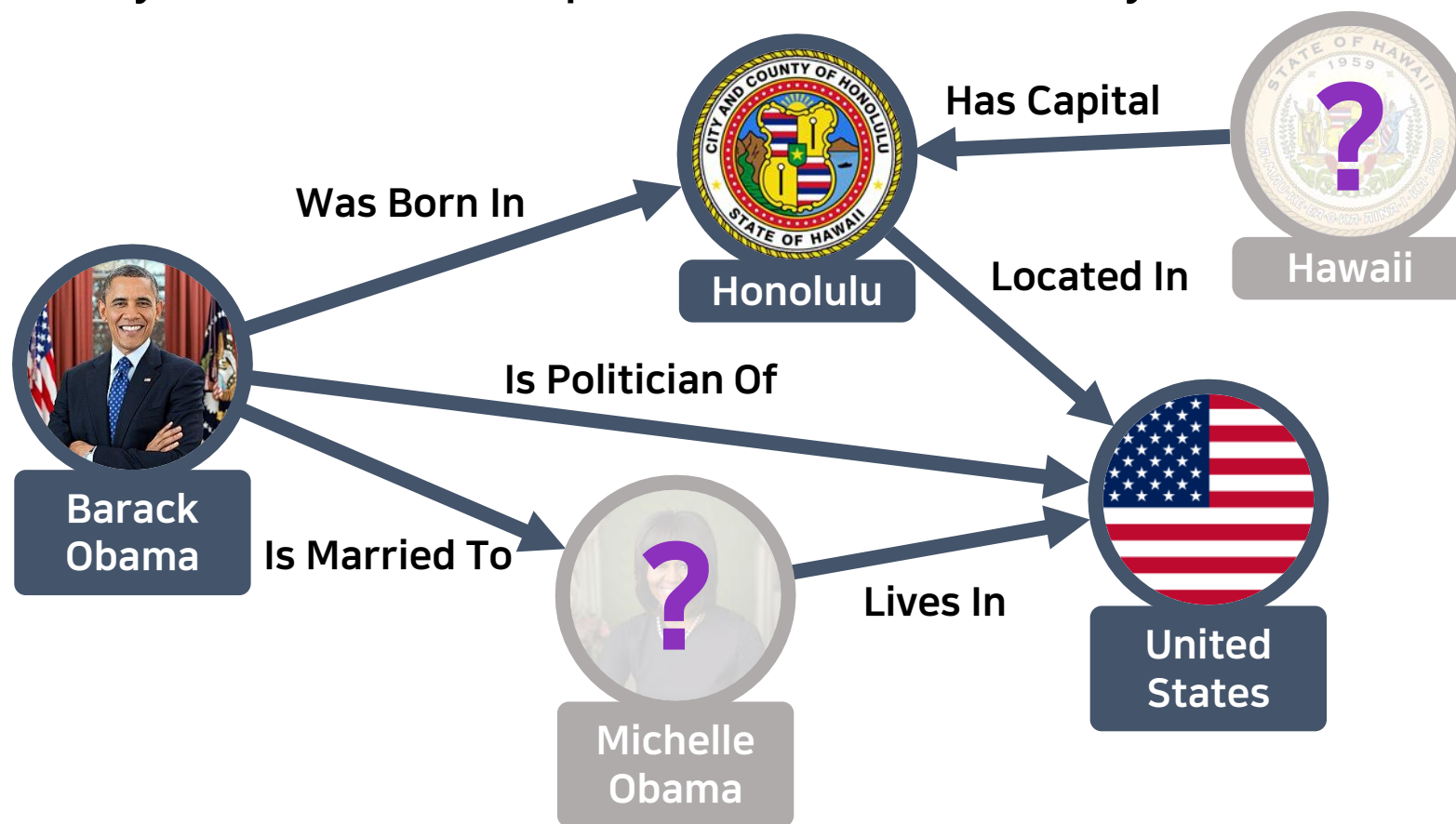
01 Knowledge Graphs

- Graphical Representation of Human Knowledge
 - Each fact is represented by a triplet (head entity, relation, tail entity)



01 Knowledge Graph Completion

- Given a head entity and a relation, predict the tail entity.
- Given a tail entity and a relation, predict the head entity.



01

Examples of Knowledge Graphs

- Google's search engine
 - An infobox next to the search results is created based on knowledge graphs

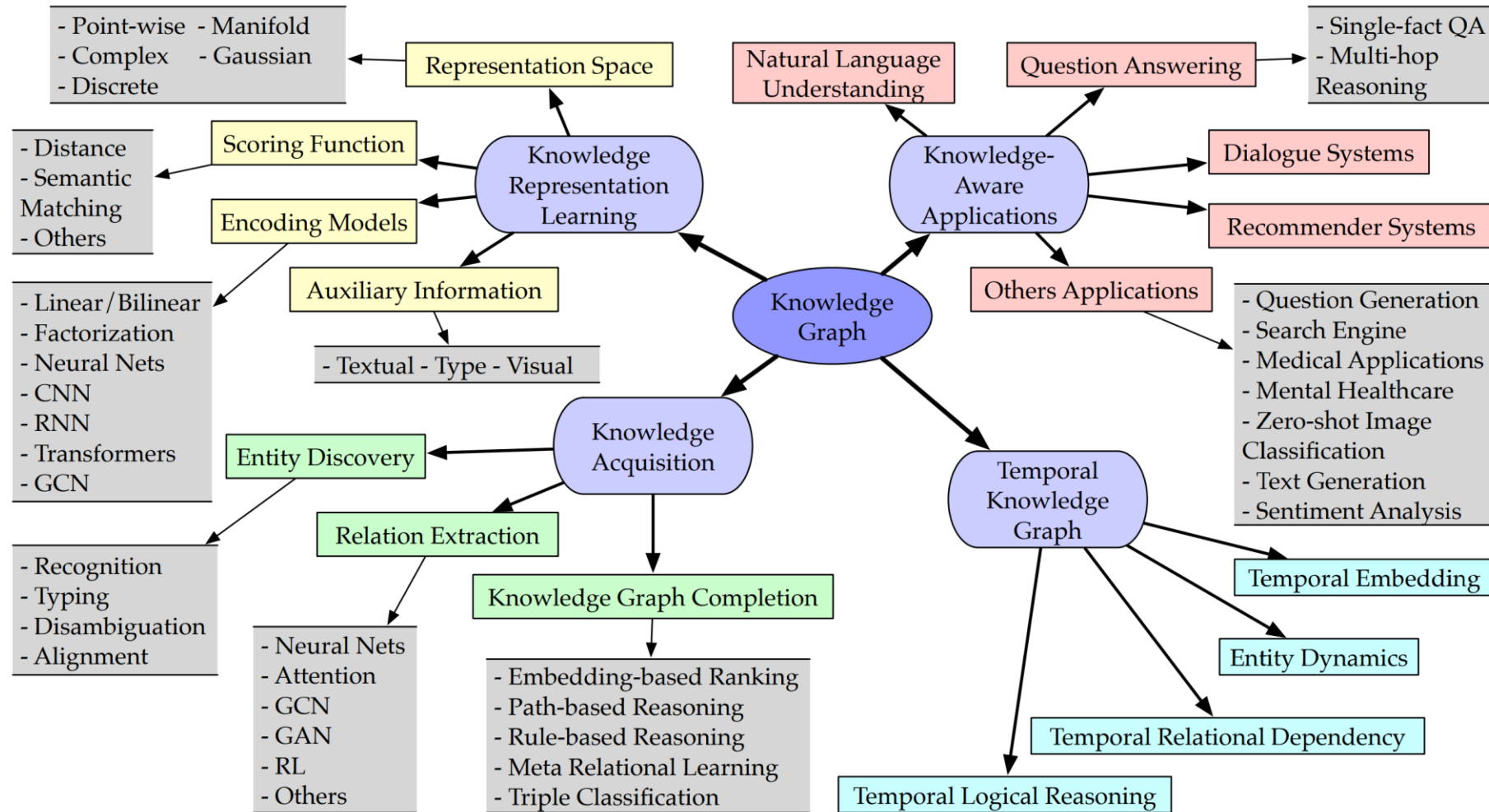
The image shows a Google search results page for the query 'kaist'. The search bar at the top contains 'kaist' and the Google logo. Below the search bar, the results show 'About 8,490,000 results (0.82 seconds)'. The first result is 'https://www.kaist.ac.kr' with the title 'KAIST'. Below the title, it says 'No information is available for this page.' and 'Learn why'. It also mentions 'You've visited this page many times. Last visit: 5/17/21'. To the right of the search results, there is a large blue-bordered box labeled 'Infobox' with an arrow pointing to it. This infobox contains a map of KAIST, a photo of the campus, and detailed information about the university. The infobox text includes: 'KAIST 한국과학기술원 대덕캠퍼스', 'University in Daejeon', 'KAIST is a national research university located in Daedeok Innopolis, Daejeon, South Korea. KAIST was established by the Korean government in 1971 as the nation's first public, research-oriented science and engineering institution. KAIST is generally known as the most prestigious university in South Korea. Wikipedia', 'Address: 291 Daehak-ro, Eoeun-dong, Yuseong-gu, Daejeon', 'Hours: Open · Closes 6PM', 'Phone: 042-350-2114', 'Academic staff: 639 (2019)', 'Total enrollment: 10,249 (2013)', 'Postgraduates: 6,738', and 'Founded: February 16, 1971'.

Examples of Knowledge Graphs

- **Google's** search engine
 - Knowledge graphs are embedded in the search engine
- **Microsoft's** knowledge mining API
 - Used for the Bing search engine, QnA pair mining, processing LinkedIn data
- **Meta's** heterogeneous graphs
 - Analyze connections between people, events, ideas, and news
- **Amazon's** product networks
 - Utilize relationships between users, products, and their metadata
- **IBM's** knowledge graphs
 - Provide a framework to develop internal knowledge graphs
- **Palantir's** ontology
 - Build business-level ontology for real-world organization management

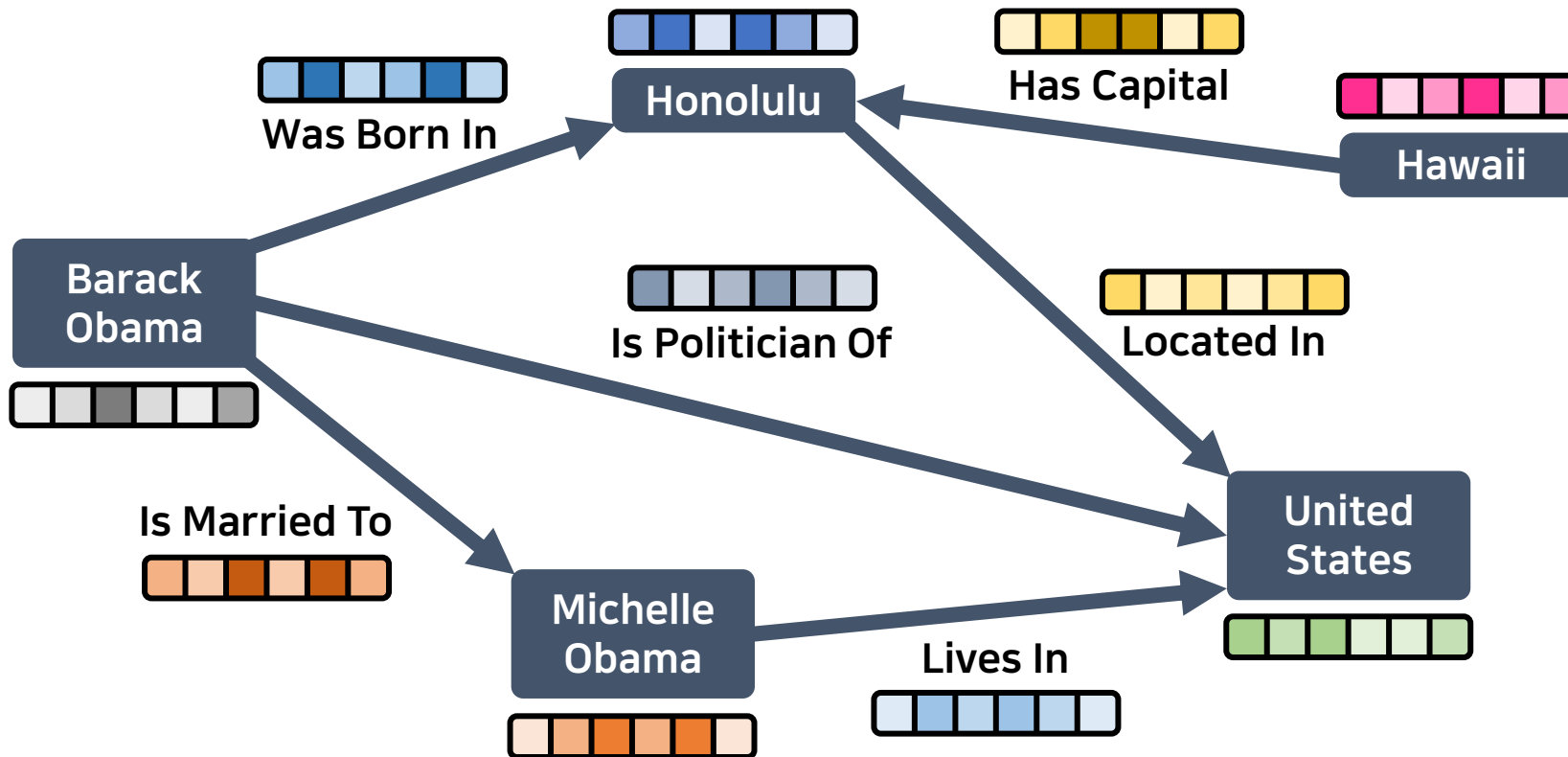


Research on Knowledge Graphs



Knowledge Graph Embedding

- Represent the entities and relations in a **low-dimensional feature space**.
- Simplify the manipulation while **preserving the inherent structure** of the graph.

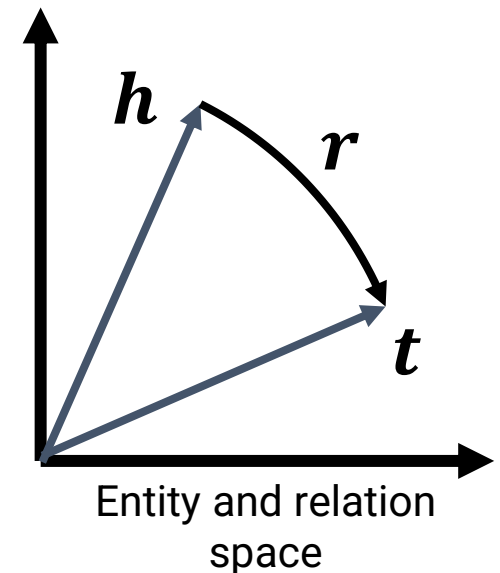
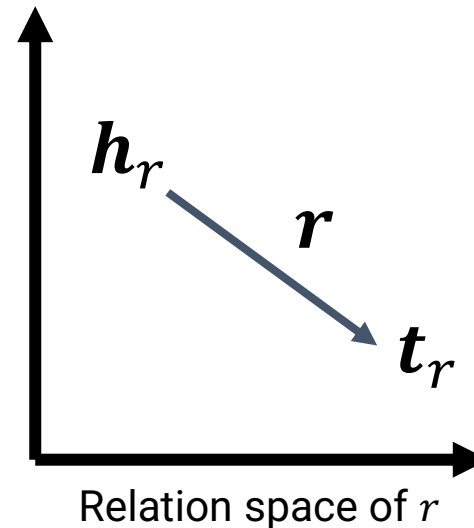
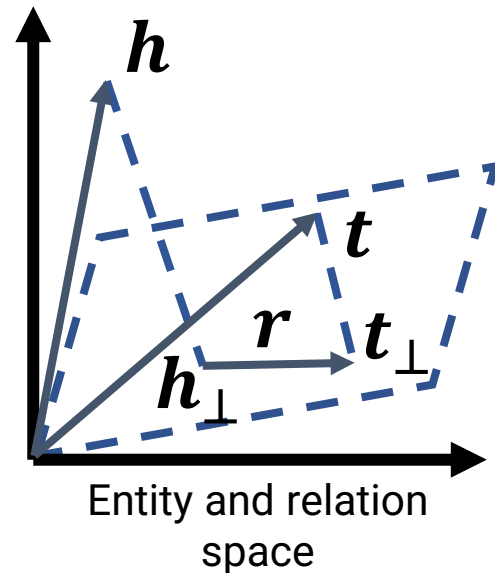
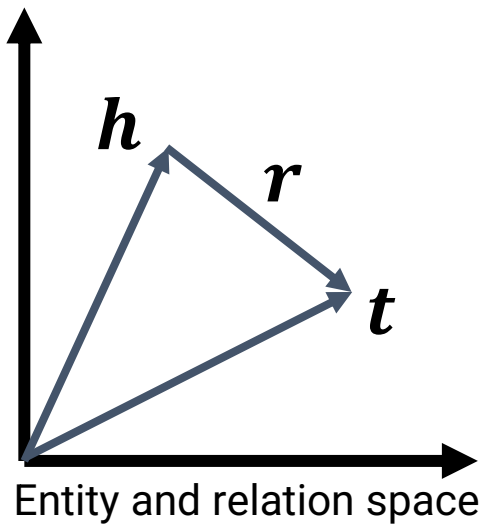


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Knowledge Graph Embedding Models

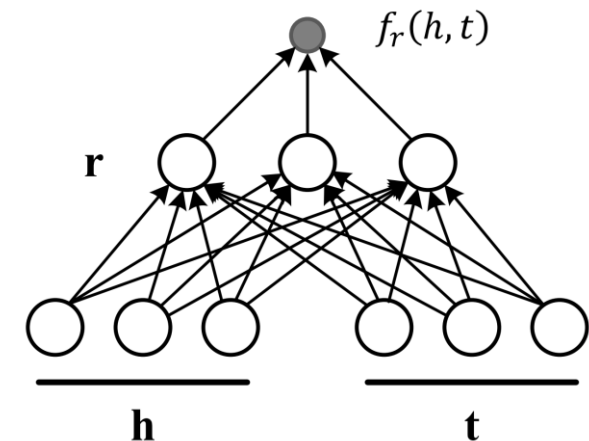
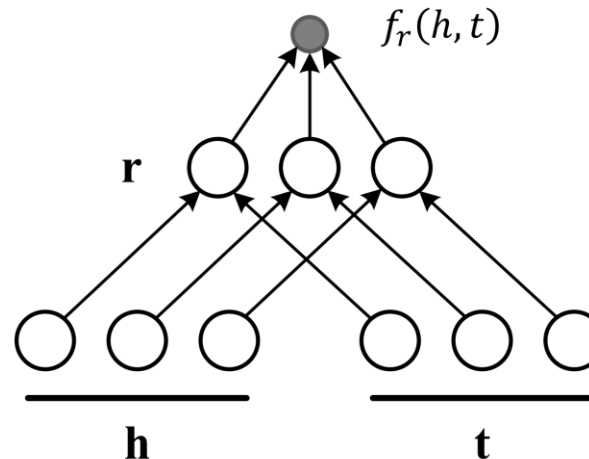
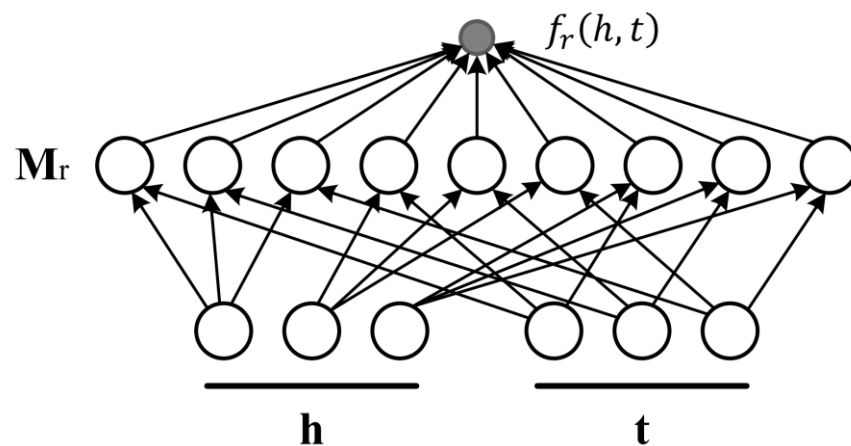
- Translational distance models

- Use distance-based scoring functions.
- Measure the plausibility of a fact as the distance between the two entities, usually after a translation carried out by the relation.
- Examples: TransE, TransH, TransR, RotatE



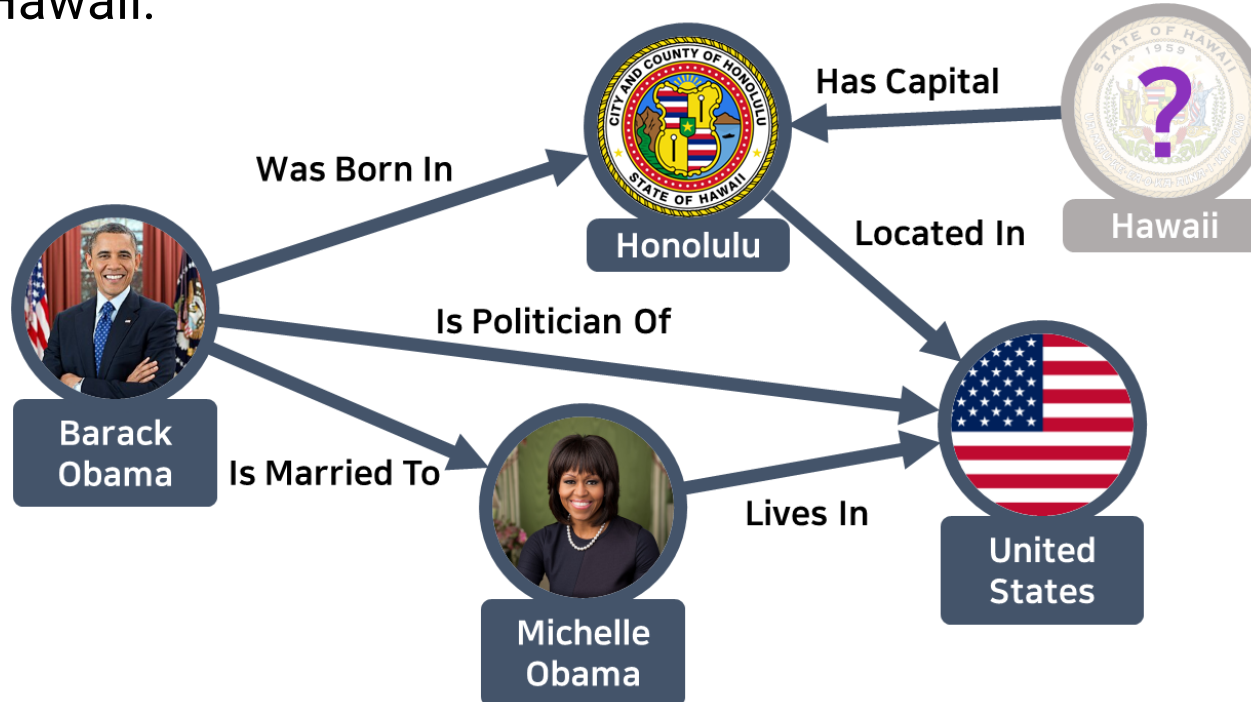
- Semantic matching models

- Exploit similarity-based scoring functions.
- Measure plausibility of facts by matching latent semantics of entities and relations embodied in their vector space representations.
- Examples: RESCAL, DistMult, HolE



01 Link Prediction in Knowledge Graphs

- Assume that **we learned embeddings** using TransE.
- To solve (?, HasCapital, Honolulu), compute $f_r(h, t) = -\|h + r - t\|$.
 - **A ranked list:** Hawaii, United States, Barack Obama, Michelle Obama
 - Predict ? to be Hawaii.



- Record **ranks of correct answers** in the ordered lists.
 - **Mean Rank (MR)**
 - Compute the average of the ranks.
 - The lower the better
 - **Mean Reciprocal Rank (MRR)**
 - The average of reciprocal ranks.
 - The higher the better
 - **Hit@N**
 - The proportion of ranks no larger than N .
 - The higher the better

Evaluation of Link Prediction

[Problem 1] Answer: D

Rank	Entity
1	B
2	C
3	D
4	A
5	E
6	F
7	H
8	J
9	K
10	G
11	I
12	L
⋮	

[Problem 2] Answer: G

Rank	Entity
1	G
2	B
3	A
4	C
5	E
6	D
7	H
8	J
9	L
10	F
11	K
12	I
⋮	

[Problem 3] Answer: K

Rank	Entity
1	A
2	C
3	F
4	B
5	J
6	D
7	E
8	G
9	H
10	K
11	I
12	L
⋮	

[Problem 4] Answer: L

Rank	Entity
1	C
2	D
3	F
4	B
5	E
6	G
7	I
8	A
9	J
10	K
11	H
12	L
⋮	

$$MR=(3+1+10+12)/4=6.5, MRR=[1/3+1+1/10+1/12]/4=0.3792, \text{Hit}@10=3/4=0.75$$

03 References

- Some slides are made based on the following references.
 - Q. Wang et al., “Knowledge graph embedding: a survey of approaches and applications”, TKDE, 2017.
 - H. Cai et al., “A Comprehensive Survey of Graph Embedding: Problems, Techniques, and Applications”, TKDE, 2018.
 - S. Ji et al., “A Survey on knowledge graphs: representation, acquisition and applications”, TNNLS, 2021.